

Ethan Lee Jun Hao

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Skills

Java, Kotlin, Python, C++, C#, SQL, Git, Android, Spring Boot, React, Docker, Linux, Unity

Projects

Commuter Android app

Built an Android app in Kotlin that shows live bus arrivals. Used activities and fragments and handled the activity lifecycle correctly when the screen rotated. Stored favourite stops on the device with a local Room database so they were available offline. Connected the app to a public transport web service and parsed its JSON responses. Published a beta build to the Play Store for classmates and fixed the crashes they reported.

Technologies: Kotlin, Android, Room

Order processing service

Implemented REST endpoints in Java using Spring Boot and dependency injection for a mock online shop. Separated the code into controllers, services and repositories. Wrote object-oriented domain classes for orders, customers and products. Added integration tests with an in-memory database. Documented the endpoints so a teammate could consume them from the front end.

Technologies: Java, Spring Boot, REST

Algorithms practice repository

Solved sixty algorithm problems in C++ and Java covering graphs, dynamic programming, heaps and string matching. Wrote short notes on the time and space complexity of each solution. Compared several sorting implementations on random and nearly sorted inputs to see how they behaved. Took part in two university programming contests.

Technologies: C++, Java

Team chat web app

Built a React front end for a team chat application with channels, message history and typing indicators. Connected the interface to a WebSocket server and a REST API for message history. Managed the work with Git branches and pull requests in a team of four. Styled the interface to work on both phones and laptops.

Technologies: React, JavaScript, Git

Personal blog engine

Wrote a small blog engine in Python with Flask that stores posts and comments in a relational database. Wrote SQL queries for tag pages and pagination. Added unit tests for the post-creation and comment rules. Deployed the blog in a Docker container on a rented Linux server and set up automatic restarts.

Technologies: Python, Flask, SQL, Docker, Linux

Game jam platformer

Programmed a small platformer in Unity with C# during a 48-hour game jam. Wrote the player movement, enemy patrol behaviour and simple collision responses using vectors. Tuned jump height and acceleration until the controls felt responsive. Shared a playable build with the other jam participants.

Technologies: Unity, C#

Experience

Software engineering intern

Worked in a team of six on a Java service that processes customer invoices. Fixed reported bugs, wrote unit tests for the modules I changed and took part in daily stand-ups and code reviews. Added a new validation rule for invoice totals and updated the related documentation.

Technologies: Java, Git

Hackathon mentor

Helped first-time participants at two weekend hackathons debug their Python and JavaScript projects. Explained version control basics and how to split a project into tasks the team could finish in a weekend.

Technologies: Python, JavaScript

Open source contributor

Fixed several small issues in an open-source command line tool written in Python, including a crash on empty input and unclear error messages. Wrote tests for each fix and responded to maintainers' review comments.

Technologies: Python, Git

Volunteer developer for a local charity

Maintained the charity's volunteer roster website, added an event calendar page and fixed layout problems on mobile phones. Trained two staff members to update the content themselves.

Technologies: HTML, CSS, JavaScript

Education

BSc Computer Science - Data structures, operating systems and software engineering modules completed